



**CURIOSITY
CREATES**

**A BETTER
WORLD**

**TOPIC 1 –
NOMENCLATURE AND TYPES OF REACTIONS**

QEng Prep - Chemistry



FACULTY OF
ENGINEERING AND
APPLIED SCIENCE

KEY CONCEPTS

The key concepts covered in this lecture include:

- Using IUPAC nomenclature conventions to identify names, write chemical formulae, and create structural formulae for the different classes of organic compounds;
 - Comparing the different classes of organic compounds by describing the similarities and differences in names and structural formulae of the compounds within each class; and
 - Explaining the chemical changes that occur during various types of organic chemical reactions, including substitution, addition, elimination, oxidation, esterification, and hydrolysis.
-

IUPAC NOMENCLATURE CONVENTION

The International Union of Pure and Applied Chemistry (IUPAC) nomenclature includes the following conventions and procedure:

Recall:

Common prefixes for alkyl groups:

List of Prefixes For The Number of Carbons Atoms In A Parent Chain:

meth-	1 carbon	hex-	6 carbons
eth-	2 carbons	hept-	7 carbons
prop-	3 carbons	oct-	8 carbons
but-	4 carbons	non-	9 carbons
pent-	5 carbons	dec-	10 carbons

Petrucci, R.H.; F.G. Herring; J.D. Madura; and C. Bissonnette (2016). *General Chemistry: Principles and Modern Applications*, Eleventh Edition, Pearson. (Chapter 26)

Step 1:

Determine the primary functional group

Step 2:

Determine the primary carbon chain

IUPAC NOMENCLATURE CONVENTION

Step 3:

Name the primary carbon chain as well as the primary functional group.

Step 4:

Number all the carbons in the primary chain

Step 5:

Indicate the position of the primary functional group in between the root of the name and the suffix.

IUPAC NOMENCLATURE CONVENTION

Step 6:

Name the alkyl groups, the alkyl halides and any other substituents, and determine their positions on the primary carbon chain using the numbering established in Step 4.

Step 7:

Write out the entire name of the molecule, including the numbers for the substituents, listed in alphabetical order; Include the multiplicative prefixes for identical substituents (dichloro-, tribromo-, etc.).

Next we will introduce various types of organic molecules and review their naming conventions following the IUPAC nomenclature conventions.

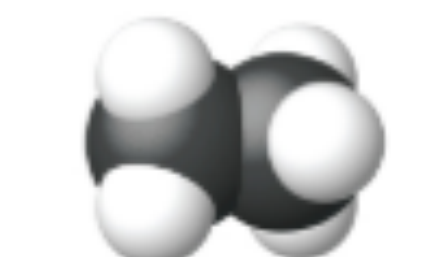
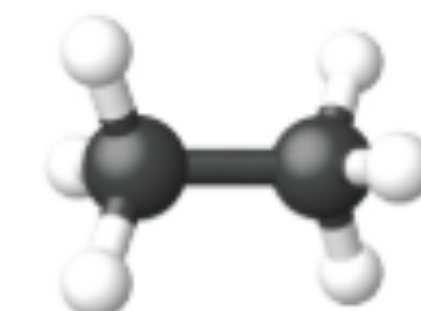
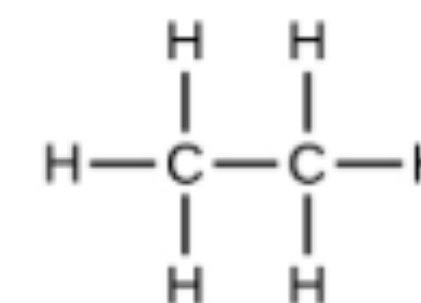
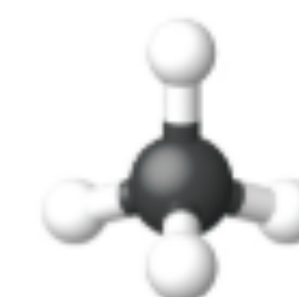
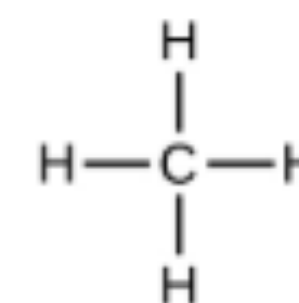
HYDROCARBONS

REVIEW: What is a hydrocarbon?

A hydrocarbon is an organic compound that consists entirely of hydrogen and carbon.

Alkanes

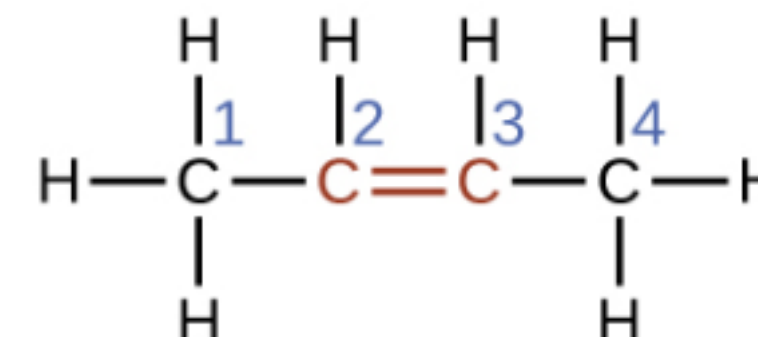
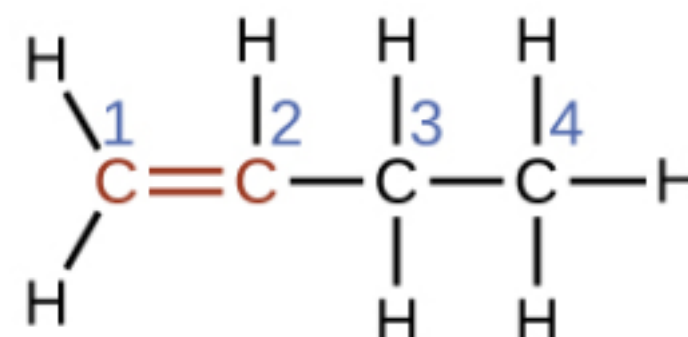
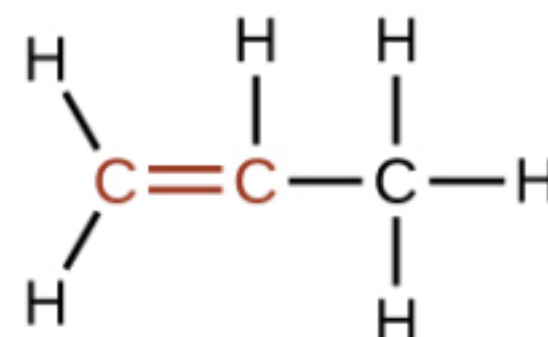
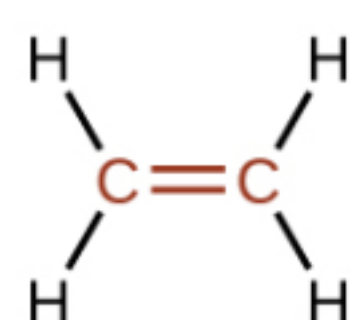
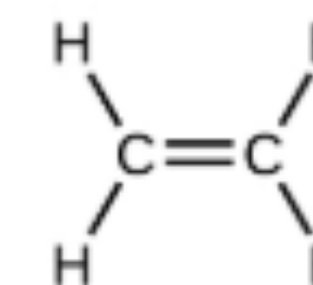
- General formula is C_nH_{2n+2}
- Saturated hydrocarbons → all carbon-carbon bonding are single bonds which are spaced tetrahedrally about carbon atoms
- For naming:
 - Use prefixes on IUPAC slide, name ends in –ane i.e., C_2H_6 is ethane
 - If there are side chains, the substituent is referred to with –yl suffixes i.e., a methane side chain is methyl, an ethane side chain is ethyl, etc.



ALKENES

Alkenes

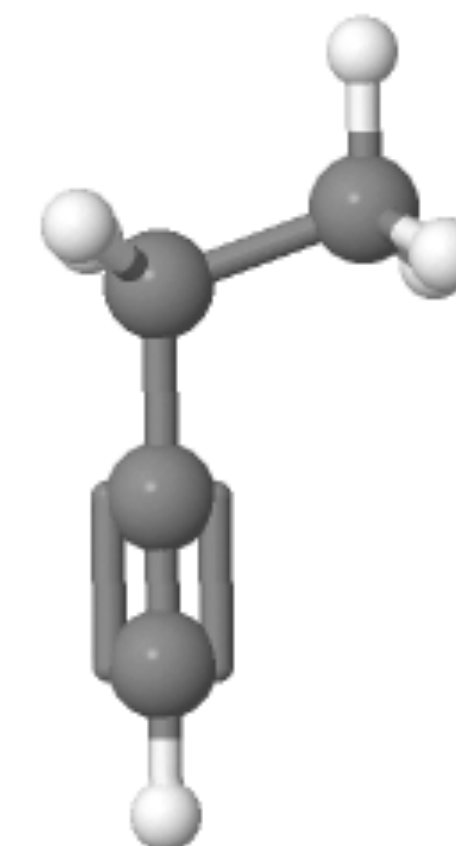
- General formula is C_nH_{2n}
- Contain a $C=C$ double bond somewhere in their structure
- Unsaturated $\rightarrow$ atoms can be added to their formula
- Spatial arrangement around the $C=C$ is planar, and the bond angles are 120°
- For naming:
 - Use prefixes on previous slide, name ends in -ene i.e., C_2H_4 is ethene



ALKYNES

Alkynes

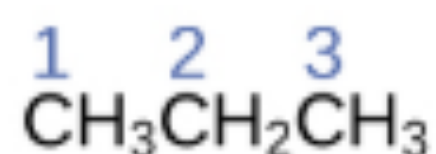
- General formula is C_nH_{2n-2}
- Unsaturated
- Contain a $C\equiv C$ triple bond somewhere in their structure
- Spatial arrangement around the $C\equiv C$ is linear, and the bond angles are 180°
- For naming:
 - Use prefixes on previous slide, name ends in -yne i.e., C_2H_2 is ethyne
 - The location of the triple bond is identified by naming the smaller of the numbers of the carbon atoms participating in the triple bond.



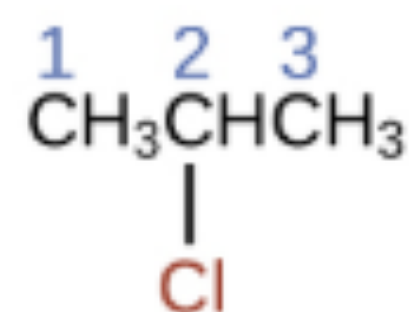
HALOGENOALKANES

Halogenoalkanes

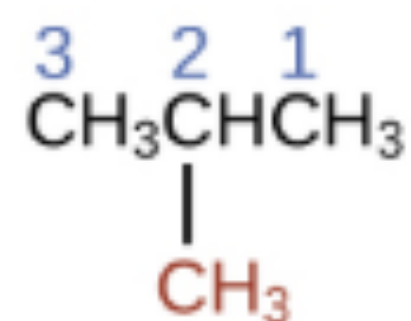
- Halogenoalkanes have the general formula $C_nH_{2n+1}X$, where X is an electronegative substituent (i.e., chloride, fluoride, bromide etc.)
- For naming:
 - The ending -ide is replaced by -o at the end of the name (i.e., chloro-, fluoro-, bromo-, etc).
 - The number of substituents of the same type is indicated by the prefixes di- (two), tri- (three), tetra- (four), and so on (for example, difluoro- indicates two fluoride substituents).



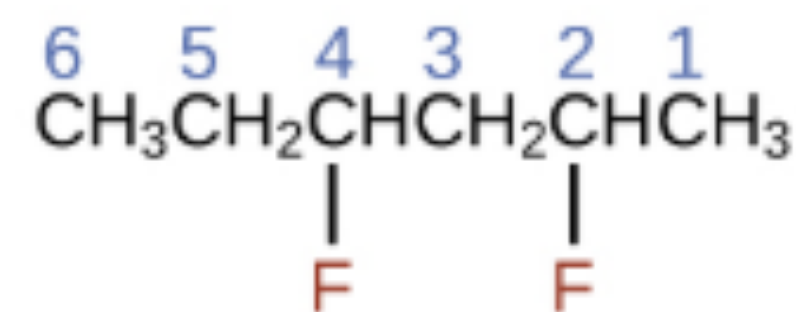
propane



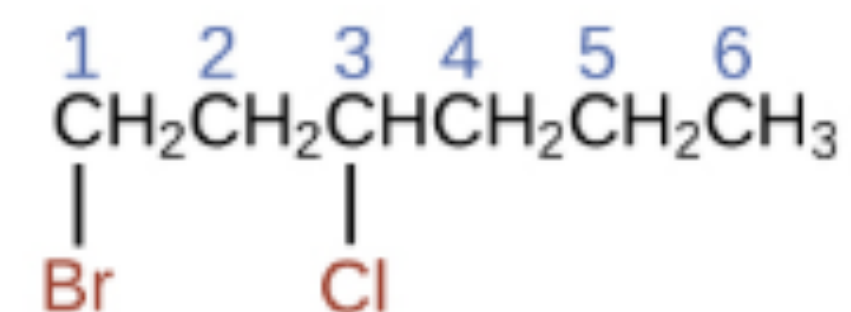
2-chloropropane



2-methylpropane



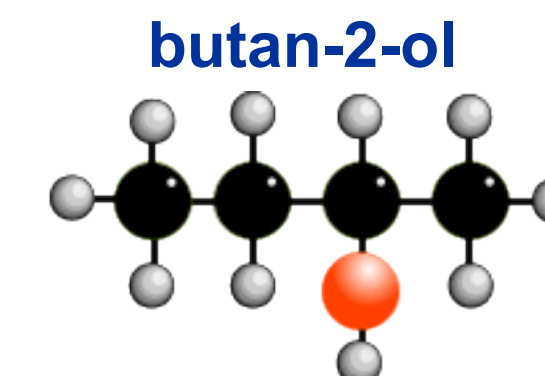
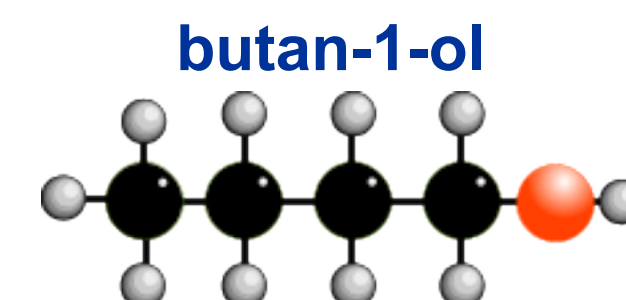
2,4-difluorohexane



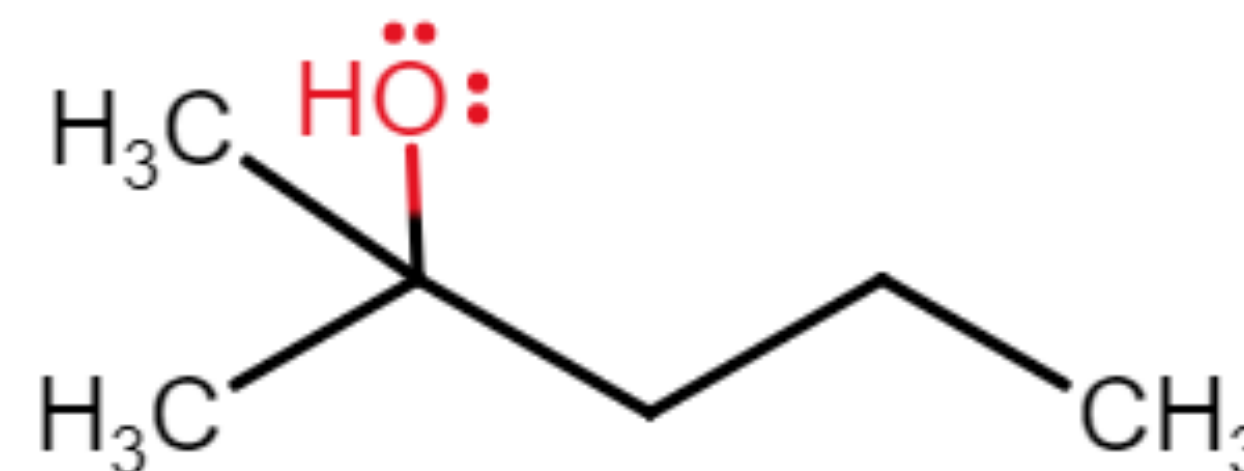
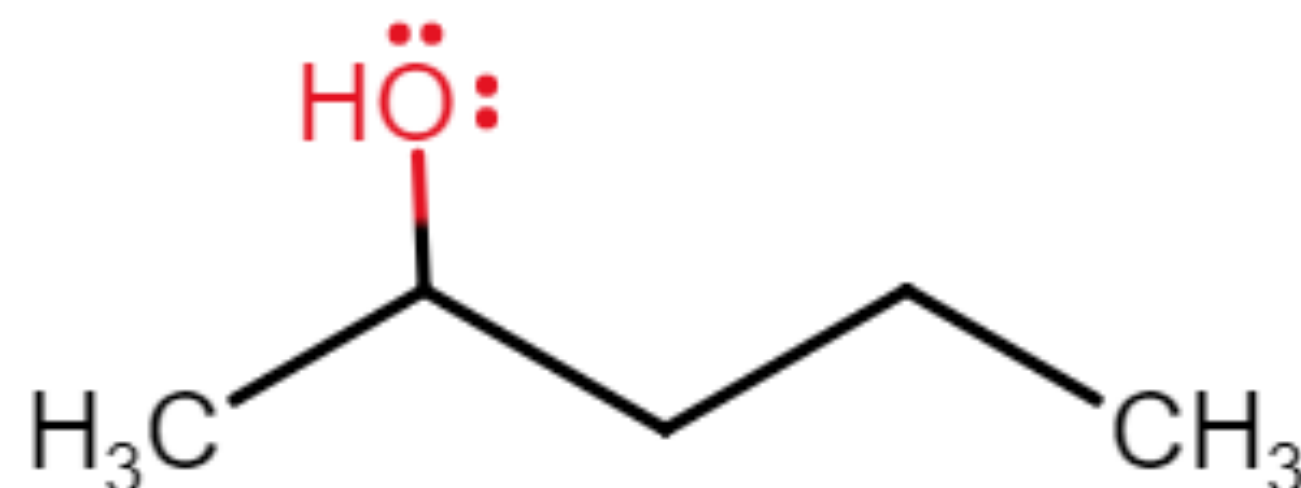
1-bromo-3-chlorohexane

ALCOHOLS

- General formula $C_nH_{2n+1}OH$
- The -OH (hydroxyl) group replaces an H in a basic hydrocarbon skeleton
- Different structures are possible with the same formula due to the position of the -OH group and the branching of the carbon chain
- For naming:
 - Select the longest chain of C atoms containing the -OH group;
 - Remove the -e at the end and add -ol after the basic name;
 - Number the chain starting from the end nearer the -OH group;
 - The number is placed after the -an and before the -ol i.e., butan-2-ol



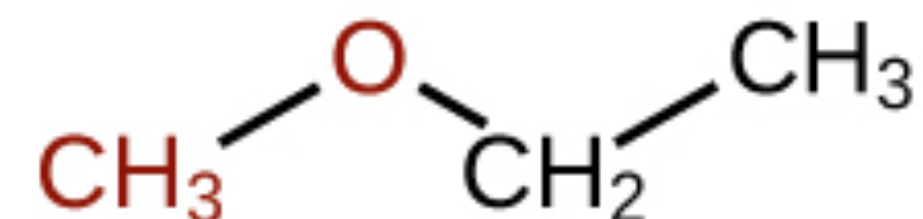
Example: What are the names of the following alcohols?



ETHERS

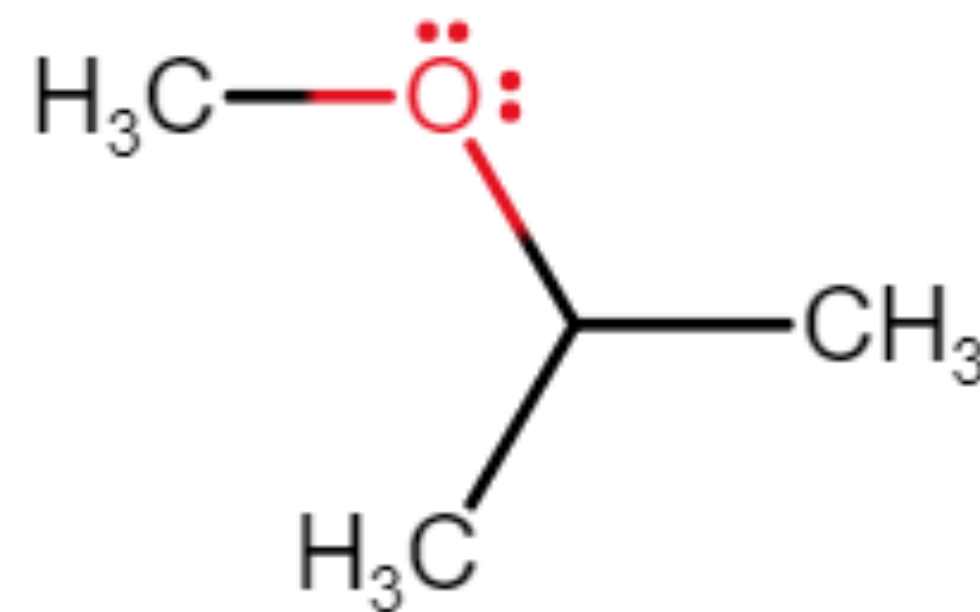
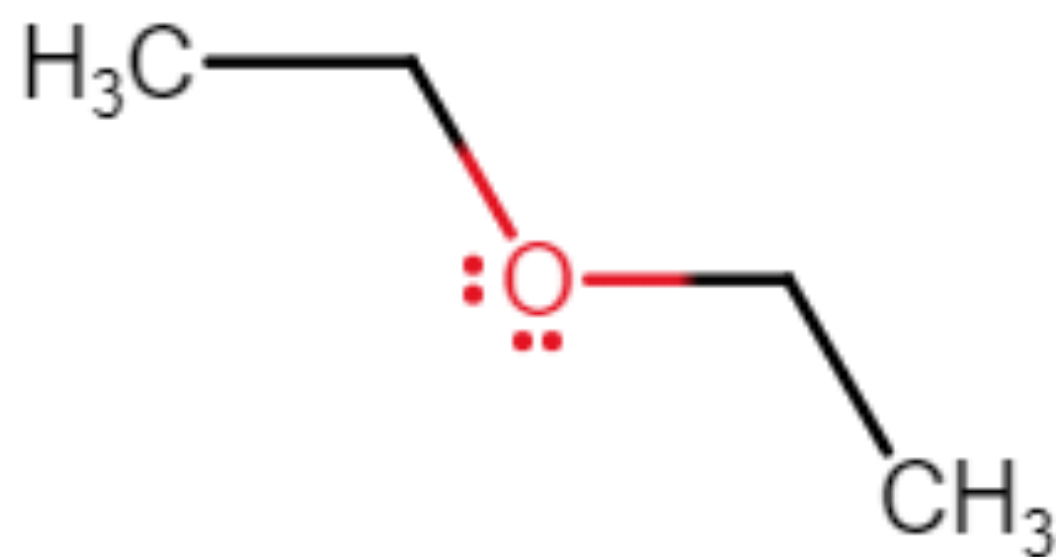
Ethers are compounds that contain the functional group --O-- .

Ethers do not have a designated suffix like the other types of molecules we have named so far. In the IUPAC system, the oxygen atom and the smaller carbon branch are named as an alkoxy substituent and the remainder of the molecule as the base chain, as in alkanes.



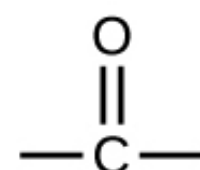
For example, with the above compound, the brown symbols represent the smaller alkyl group and the oxygen atom, which would be named “methoxy” and the larger carbon branch would be ethane, making the molecule methoxyethane.

Example: What are the names of the following ethers?

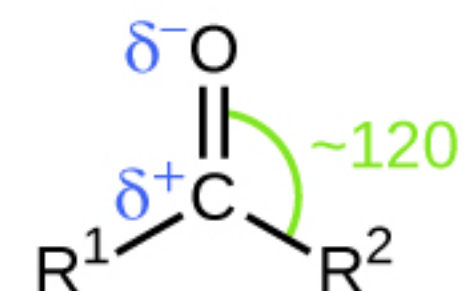


ALDEHYDES & KETONES

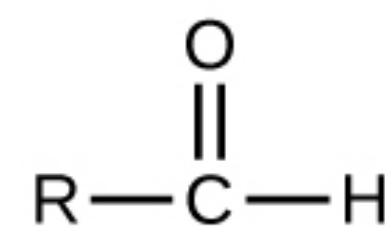
Both **aldehydes** and **ketones** contain a carbonyl group - a functional group with a carbon-oxygen double bond:



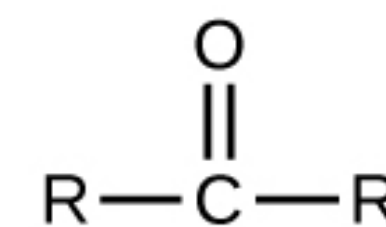
In both aldehydes and ketones, the geometry around the carbon atom in the carbonyl group is trigonal planar:



The names for aldehyde and ketone compounds are derived using similar nomenclature rules as for alkanes and alcohols, and include the class-identifying suffixes -al and -one, respectively.

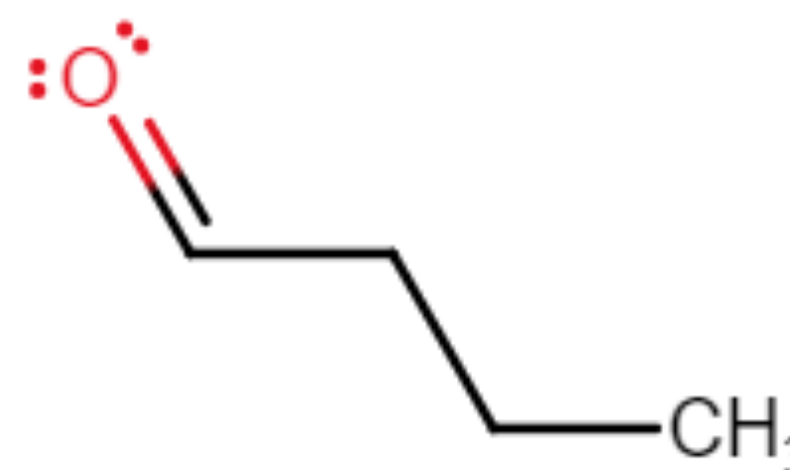
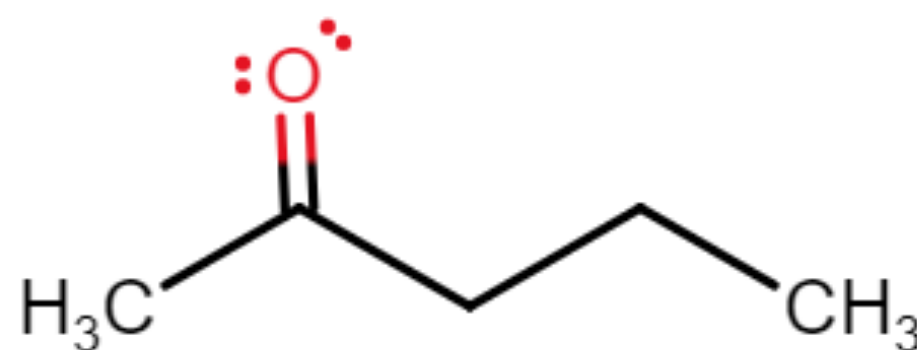


Functional group
of an aldehyde



Functional group
of a ketone

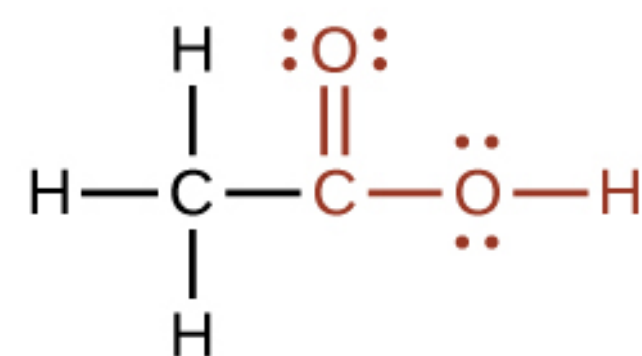
Example: What are names of the following aldehydes and ketones?



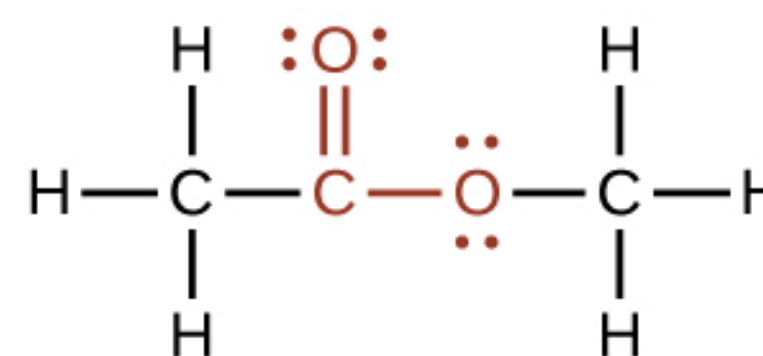
CARBOXYLIC ACIDS & ESTERS

Both **carboxylic acids** and **esters** contain a carbonyl group (a C=O group) with a second oxygen atom bonded to the carbon atom in the carbonyl group by a single bond. In a carboxylic acid, the second oxygen atom also bonds to a hydrogen atom. In an ester, the second oxygen atom bonds to another carbon atom. The functional groups are shown in brown in the image below.

The names for carboxylic acids and esters include prefixes that denote the lengths of the carbon chains in the molecules and include the class-identifying suffixes -anoic and -oate, respectively. For esters, the name identifies the carbon group bonded to the single-bonded oxygen first (i.e., methyl in the example below)

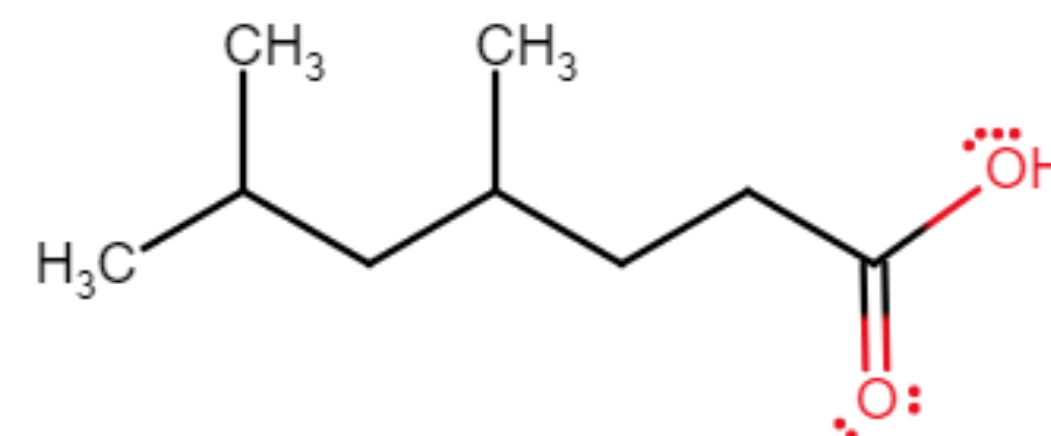
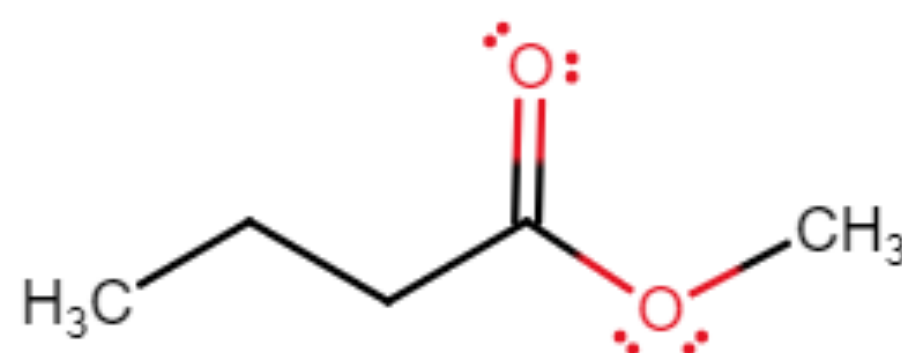


ethanoic acid



methyl ethanoate

Example: What are names of the following organic compounds?

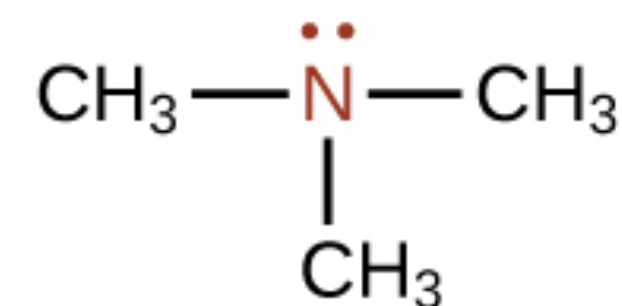
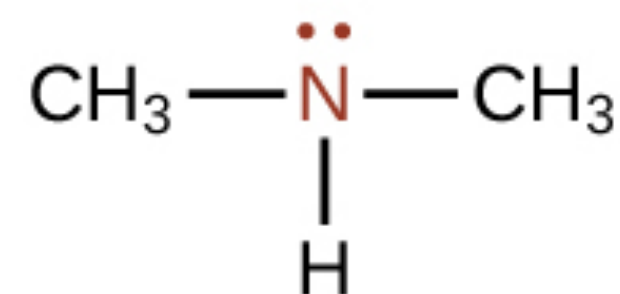
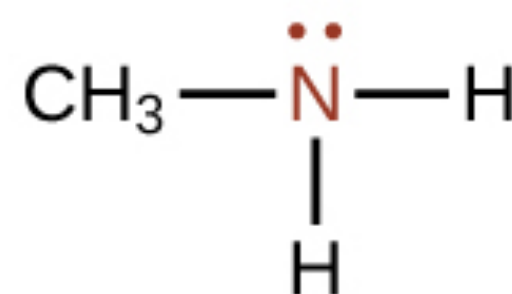


AMINES

Amines are molecules that contain C-N bonds. Amines are derivatives of ammonia in which one or more of the hydrogens has been replaced by an alkyl or aryl group.

The terms primary, secondary and tertiary are used to classify amines and refer to the number of alkyl substituents are bonded to the N atom (1, 2 and 3 alkyl substituents, respectively). The N atom in an amine has a lone pair of electrons and three bonds to other atoms, either C or H.

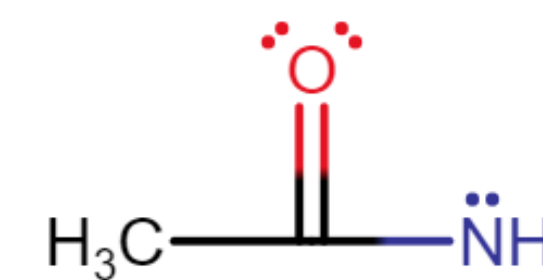
Nomenclature rules involve the class-identifying suffix –amine as illustrated here in a few simple examples:



AMIDES

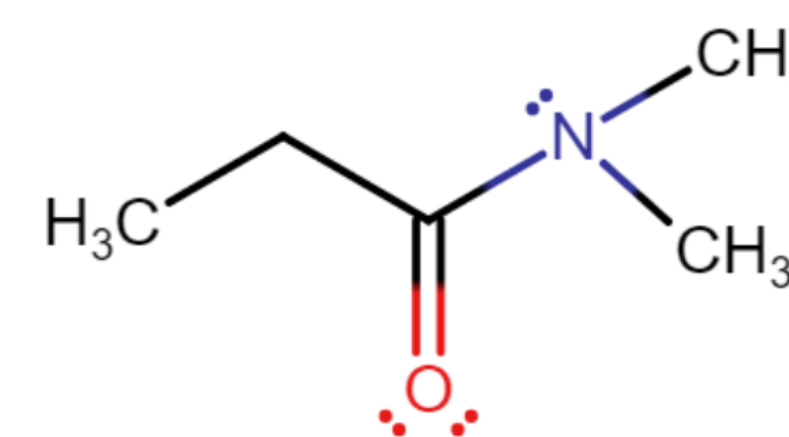
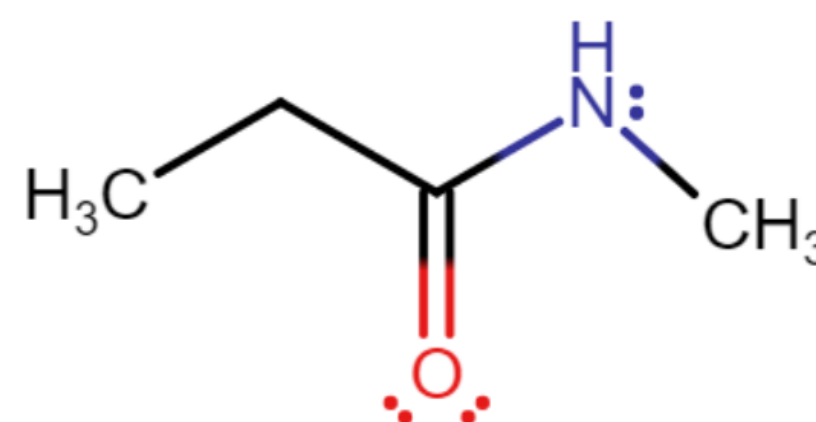
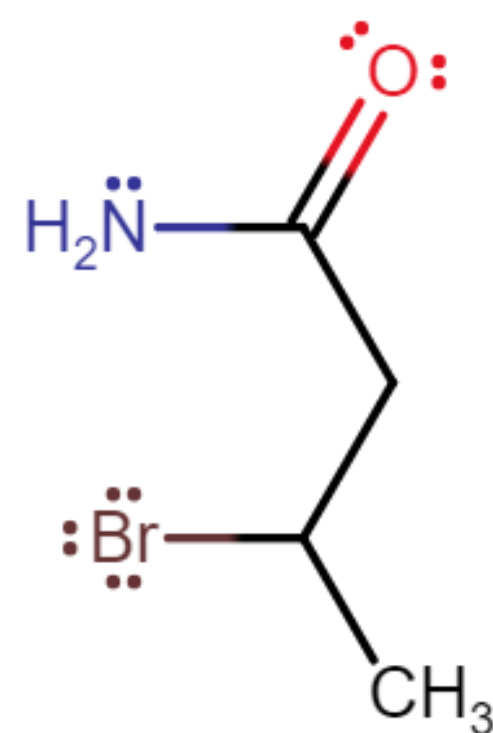
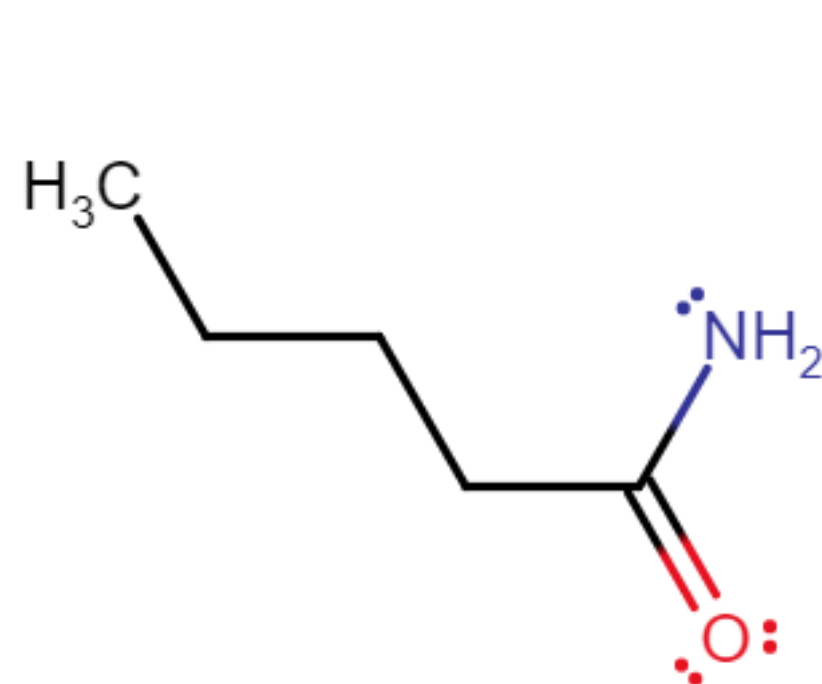
Amides are molecules that contain N atoms connected to the C atom of a carbonyl group (C=O).

Simple (*primary*) amides are named as derivatives of carboxylic acids, where the –oic ending of the carboxylic acid is replaced with the suffix –amide. For example, the molecule shown to the right is ethanamide, from the carboxylic acid ethanoic acid:



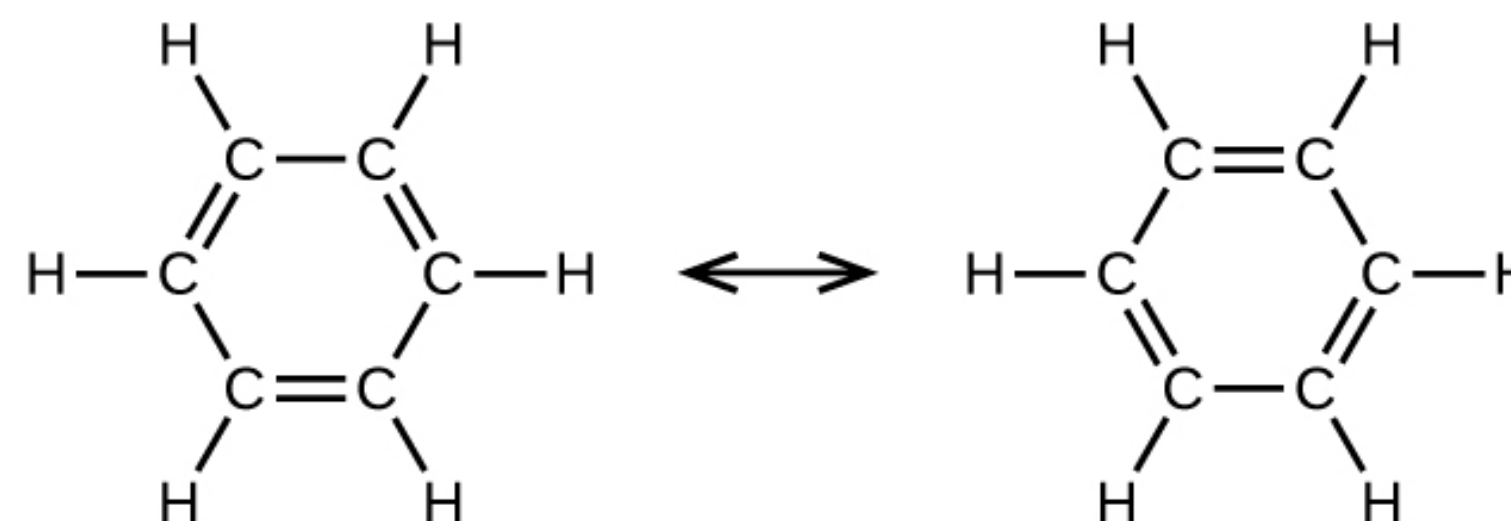
Secondary amides are named by using an upper case N to designate that there is an alkyl group attached to the N atom. Alkyl groups attached to the nitrogen are named as substituents, and the letter N is used to indicate they are attached to the nitrogen.

Example: What are names of the following organic compounds?

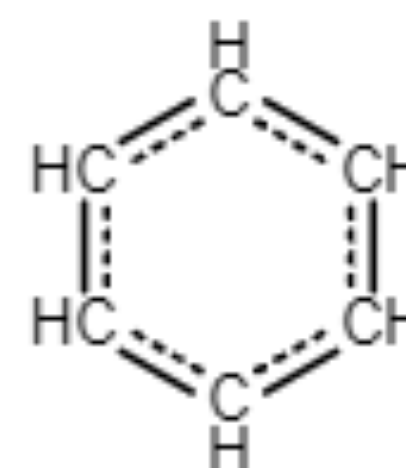


AROMATIC COMPOUNDS

Benzene, C_6H_6 , is the simplest member of a large family of hydrocarbons, called **aromatic hydrocarbons**. These compounds contain ring structures where carbon-carbon double bonds are shared between alternating carbons. There are two resonance structures for benzene:



However, each of the six bonds between its carbon atoms is actually equivalent and exhibits properties that are intermediate between those of a C–C single bond and a C=C double bond. To represent this unique bonding, structural formulas for benzene and its derivatives are typically drawn with single bonds between the carbon atoms and a circle within the ring as shown below:

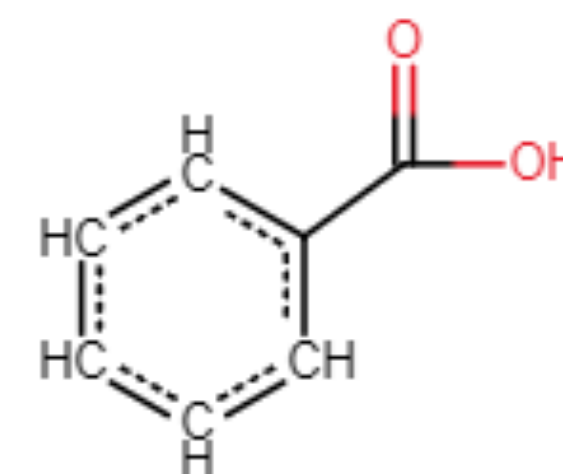
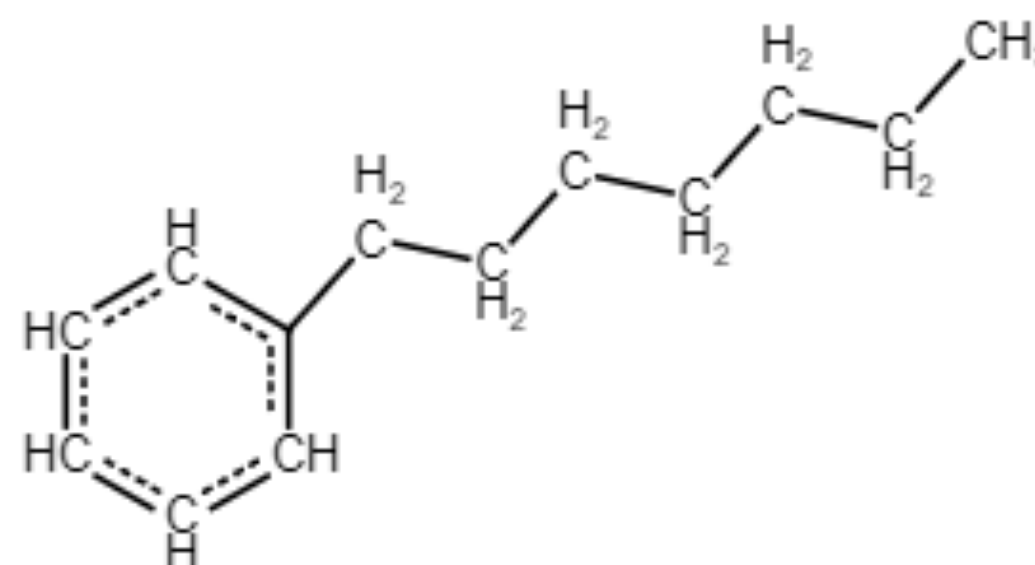
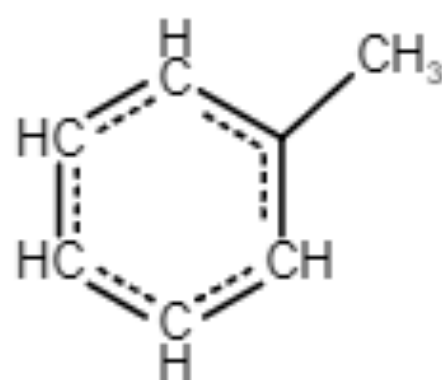


AROMATIC COMPOUNDS

Aromatic Compounds with a Single Substituent

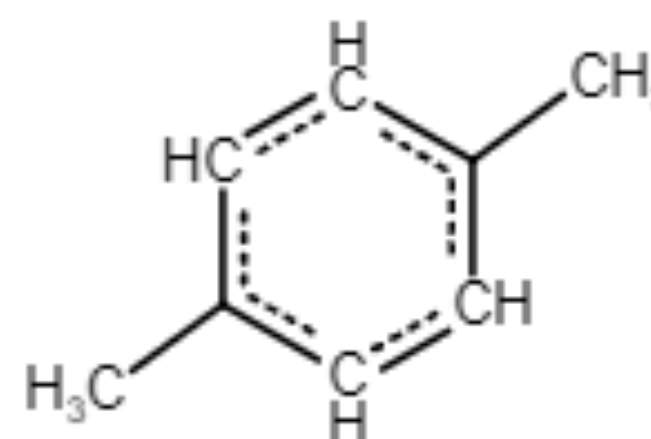
When there is a single substituent on a benzene ring and the substituent contains six or fewer carbons, the substituent is included as a prefix to benzene. Alkyl groups are named according to the alkane series convention ending with -yl: methyl (for a single carbon), ethyl (for two carbons), propyl (for three carbons), etc.

If the substituent contains more than six carbons, the alkane portion is named first, and the aromatic ring portion is added as a suffix as **phenyl-**. For example, methylbenzene is shown below on the left, 1-phenyloctane is shown in the middle, and benzoic acid is shown on the right:

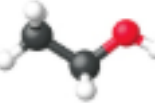
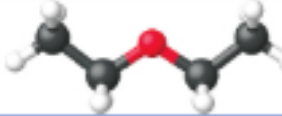
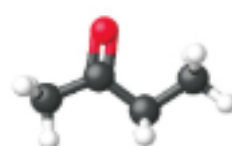
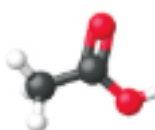
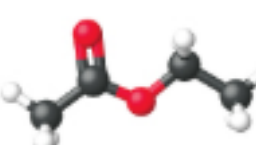
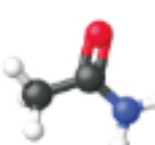


Aromatic Compounds with Multiple Substituents

When there are multiple substituents, ring atoms are numbered to minimize the numbers assigned to the substituted positions, similar to a traditional alkane or alkene. For example, 1,4-dimethylbenzene is shown below:



SUMMARY

Compound Name	Structure of Compound and Functional Group (red)	Example	
		Formula	Name
alkene	$C=C$	C_2H_4 	ethene
alkyne	$C\equiv C$	C_2H_2 	ethyne
alcohol	$R-\ddot{O}-H$	CH_3CH_2OH 	ethanol
ether	$R-\ddot{O}-R'$	$(C_2H_5)_2O$ 	diethyl ether
aldehyde	$\begin{array}{c} :\ddot{O}: \\ \\ R-C-H \end{array}$	CH_3CHO 	ethanal
ketone	$\begin{array}{c} :\ddot{O}: \\ \\ R-C-R' \end{array}$	$CH_3COCH_2CH_3$ 	methyl ethyl ketone
carboxylic acid	$\begin{array}{c} :\ddot{O}: \\ \\ R-C-\ddot{O}-H \end{array}$	CH_3COOH 	acetic acid
ester	$\begin{array}{c} :\ddot{O}: \\ \\ R-C-\ddot{O}-R' \end{array}$	$CH_3CO_2CH_2CH_3$ 	ethyl acetate
amine	$\begin{array}{c} \ddot{N}-H \\ \\ H \end{array} \quad \begin{array}{c} \ddot{N}-H \\ \\ R' \end{array} \quad \begin{array}{c} \ddot{N}-R'' \\ \\ R' \end{array}$	$C_2H_5NH_2$ 	ethylamine
amide	$\begin{array}{c} :\ddot{O}: \\ \\ R-C-\ddot{N}-R' \\ \\ H \end{array}$	CH_3CONH_2 	acetamide

Priority	Functional Group
1	Carboxylic Acid
2	Ester
3	Amide
4	Aldehyde
5	Ketone
6	Alcohol
7	Amine
8	Alkene
9	Alkyne
10	Alkane

TYPES OF CHEMICAL REACTIONS

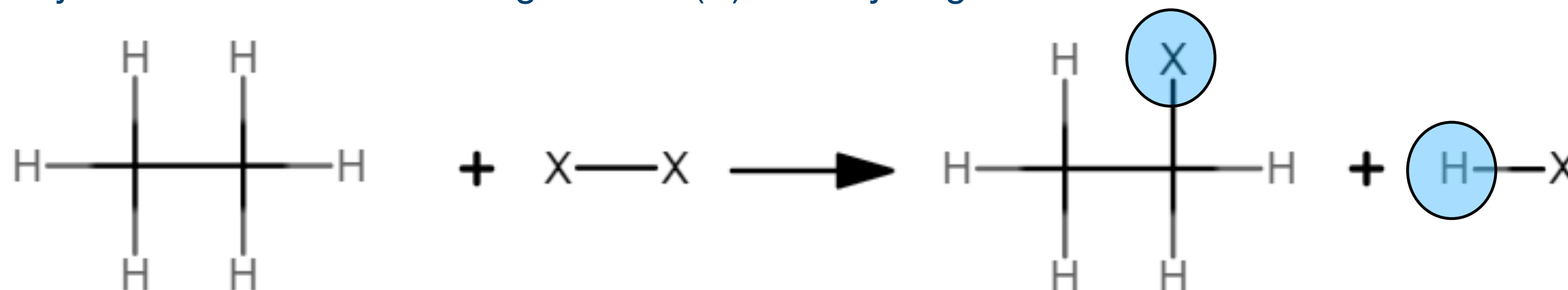
SUBSTITUTION REACTIONS

Definition

A substitution reaction is a reaction in which one or more atoms in a molecule are replaced with another atom or group of atoms.

Examples

1. Alkyl halides are formed by the substitution of a halogen atom (X) for a hydrogen atom:



2. When methane reacts with chlorine gas, ultraviolet light can act as a catalyst for the reaction:



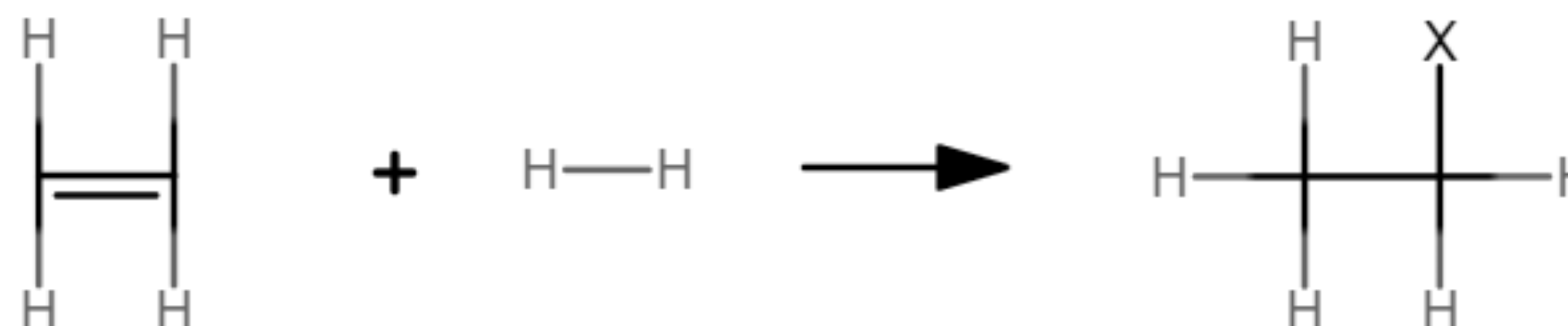
ADDITION REACTIONS

Definition

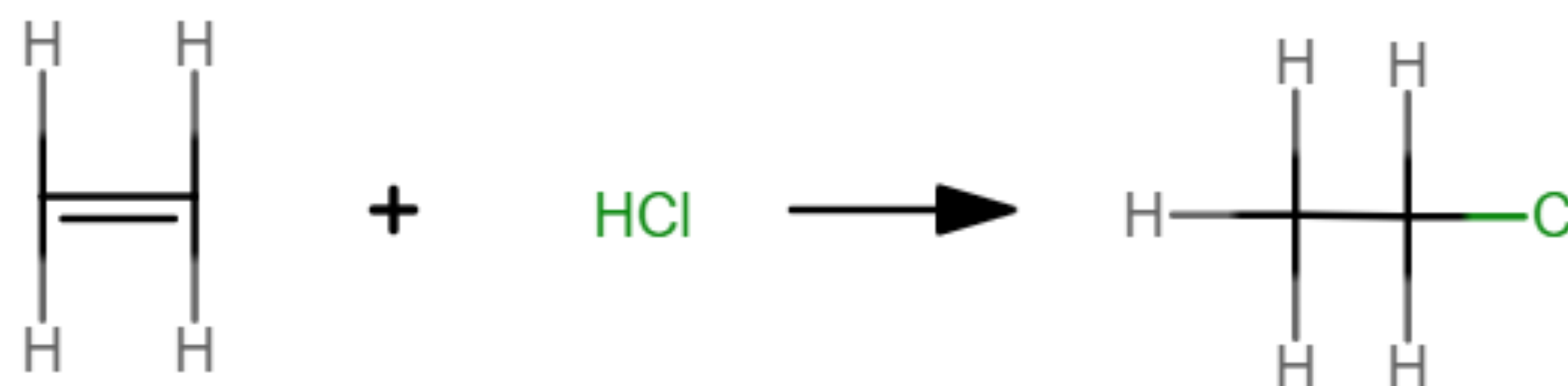
An addition reaction is a reaction in which an atom or molecule is added to an unsaturated molecule (double or triple bond), making a single product.

Examples

1. Hydrogenation is a reaction that occurs when molecular hydrogen is added to an alkene to produce an alkane or hydrogen is added to an alkyne to produce an alkene or alkane. For example, ethene reacts with hydrogen to form ethane, with the use of a transition metal catalyst.



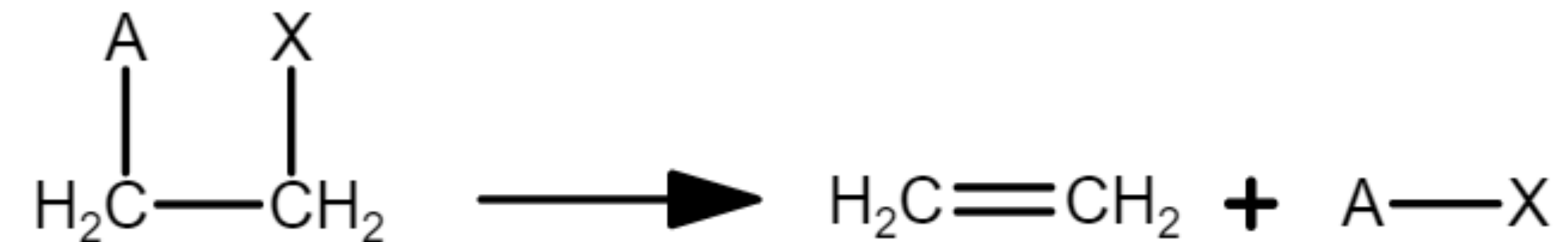
2. Addition reactions are also useful ways to introduce a new functional group into an organic molecule. Alkyl halides can be produced from an alkene by the addition of either the elemental halogen or the hydrogen halide. A monosubstituted alkyl halide can be produced by the addition of a hydrogen halide to an alkene. Shown below is the formation of chloroethane.



ELIMINATION REACTIONS

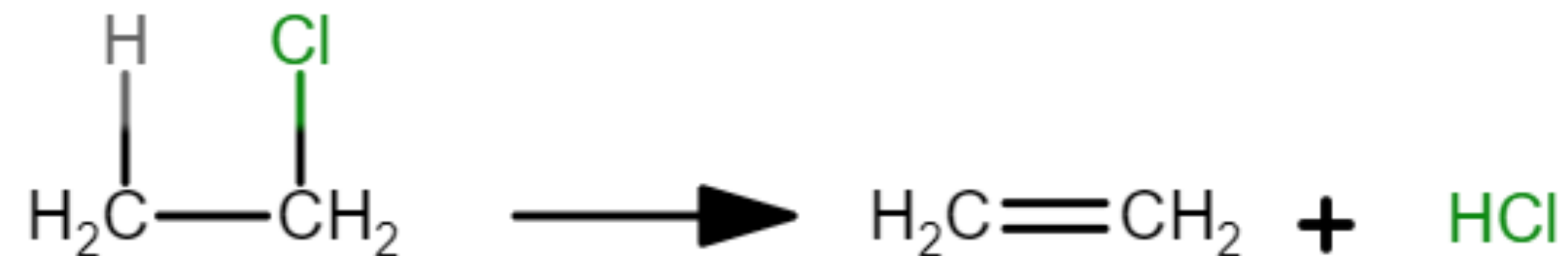
Definition

An elimination reaction involves the removal of adjacent atoms from a molecule. This results in the formation of a multiple bond and the release of a small molecule, so they are called elimination reactions. They have the general form:



Example

A common elimination reaction is the conversion of chloroethane to ethylene:



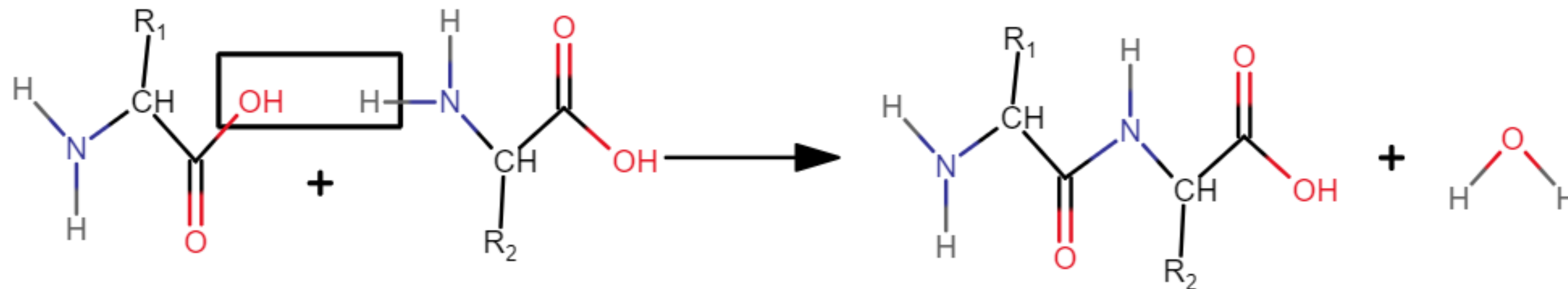
CONDENSATION REACTIONS

Definition

A condensation reaction is a reaction in which two molecules combine to form a single molecule. A small molecule, often water, is removed during a condensation reaction.

Example

Amino acids are important biological molecules that have an amine functional group on one end of the molecule and a carboxylic acid functional group on the other end. When two amino acids combine in a condensation reaction, a covalent bond forms between the amine nitrogen of one amino acid and the carboxyl carbon of the second amino acid. A molecule of water is then removed as a second product:



ESTERIFICATION

Definition

esterification is a condensation reaction in which an ester is formed from an alcohol and a carboxylic acid. Esterification is a subcategory of condensation reactions because a water molecule is produced in the reaction. The reaction can be catalyzed by a strong acid, usually sulfuric acid.

Example

When a carboxylic acid, butanoic acid, is heated with an excess of methanol and a few drops of sulfuric acid, the ester methyl butanoate is produced, which has the scent of pineapples:



HYDROLYSIS REACTIONS

Definition

A hydrolysis reaction is the reverse of a condensation reaction; "hydro" indicates that water is involved, and "lysis" means to break apart. Therefore, in a hydrolysis reaction, water is added and the molecule breaks apart, usually at a C–O–C linkage.

Example

The hydrolysis of methyl ethanoate to form ethanoic acid and methanol:

